

III.

$$(\Leftarrow) \quad r(A+B) = r(A) + r(B)$$

$$\Rightarrow C(A+B) = C(A) \oplus C(B), \quad R(A+B) = R(A) \oplus R(B)$$

$$\begin{aligned} CC^T C &= AC^T A + AC^T B + B C^T A + B C^T B \\ &= A+B \end{aligned}$$

$$A = AC^T A + AC^T B \in C(A),$$

$$B = BC^T A + BC^T B \in C(B)$$

$$R(B) \ni AC^T B = A - AC^T A = (I - AC^T)A \in R(A)$$

$$R(A) \ni BC^T A = B - BC^T B = (I - BC^T)B \in R(B)$$

$$\Rightarrow AC^T B = BC^T A = 0 \quad \because R(A) \cap R(B) = \{0\}$$

$$\Rightarrow AC^T A = A, \quad BC^T B = B$$

$$(\Rightarrow) \quad AC^T A = A, \quad BC^T B = B, \quad AC^T B = 0, \quad BC^T A = 0$$

$$\text{By hint, } r(A+B) \leq r(A) + r(B)$$

$$\text{WTS: } r(A) + r(B) \leq r(A+B)$$

CC^- : idempotent $\because CC^-CC^- = CC^-$

$$r(CC^-) \leq r(C)$$

$$\stackrel{\parallel}{=} r(CC^-)$$

$$\stackrel{\parallel}{=} r(AC^-) + r(BC^-) \quad \& \quad AC^-, BC^- \text{ idempotent.}$$

$$\stackrel{\parallel}{=} r(AC^-) + r(BC^-) \quad (\because AC^-AC^- = AC^-)$$

$$r(ACA) \leq r(AC^-) \quad \& \quad r(BC^-B) \leq r(BC^-)$$

$$\Rightarrow r(A) + r(B) \leq r(C)$$

□

2.

$$\frac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma \sqrt{n^{-1}}} \sim N(0, 1)$$

$$\frac{\hat{n} S_p^2}{\sigma^2} = \frac{(n_1 - 1) S_1^2}{\sigma^2} + \frac{(n_2 - 1) S_2^2}{\sigma^2} = \chi^2(n_1 - 1) + \chi^2(n_2 - 1) \\ \sim \chi^2(n_1 + n_2 - 2)$$

$$\Rightarrow \frac{\hat{n} S_p^2}{\sigma^2} / \hat{n} \sim \chi^2(n_1 + n_2 - 2) / \hat{n}$$

$$\bar{y}_1, \bar{y}_2 \perp S_1^2, S_2^2 \Rightarrow \bar{y}_1 - \bar{y}_2 \perp S_p^2$$

$$t = \frac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma \sqrt{n^{-1}}}$$

$$\frac{t}{\sqrt{\frac{\hat{n} S_p^2}{\sigma^2} / \hat{n}}} = \frac{Z}{\sqrt{\chi^2(\hat{n}) / \hat{n}}} \\ \sim t(n_1 + n_2 - 2)$$

$$t^2 \sim F(1, \hat{n})$$

□

[3].

$$\begin{aligned}(\Leftarrow) \quad & X^T X (\hat{\beta} + (I - \bar{X} \bar{X})z) \\&= X^T X \hat{\beta} + (X^T X - \underbrace{X^T X \bar{X} \bar{X}}_{=X})z \\&= \bar{X}^T y : \text{solution.}\end{aligned}$$

($\Rightarrow$) Note that the solution of $X^T X \beta = X^T y$ can be expressed by $\hat{\beta} = x_p + x_h$, where x_p is a solution of $X^T X \beta = X^T y$, and x_h is a solution of $X^T X \beta = 0$. Here, $x_h \in N(X^T X)$

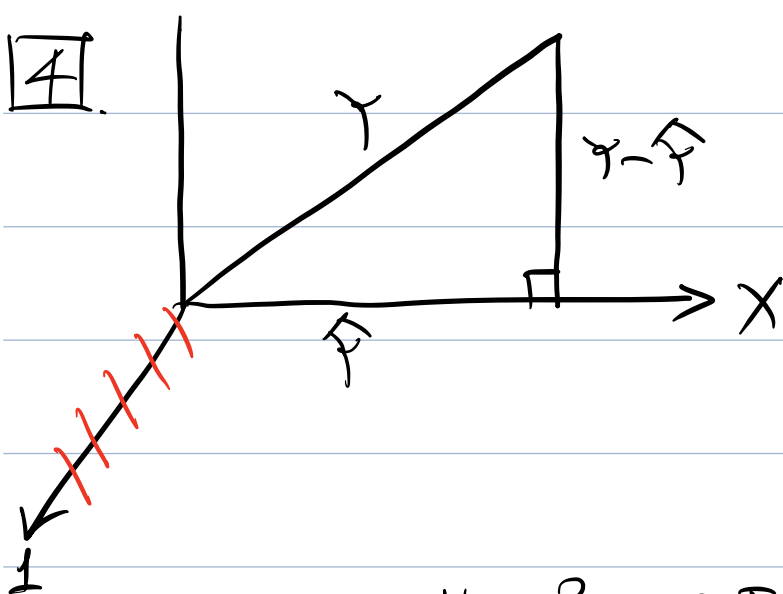
$$\& X^T X (I - \bar{X} \bar{X}) = X^T X - X^T X = 0$$

$$\text{So } X^T X (I - \bar{X} \bar{X})z = 0 \Rightarrow (I - \bar{X} \bar{X})z \in N(X^T X).$$

Also, a possible x_p is $\hat{\beta} = \bar{X}^T y$ in (1)

$$\begin{aligned}\text{So, } \hat{\beta} &= x_p + x_h \\&= \bar{X}^T y + (I - \bar{X} \bar{X})z\end{aligned}$$

□



(a) $SST = \|Y\|^2 = Y^T Y$

$$SSE = \|Y - \hat{Y}\|^2 = Y^T (I - H_X) Y$$

$$SSR = \|\hat{Y}\|^2 = Y^T H_X Y$$

∴ check: $SST = SSE + SSR$

$$Y^T Y = Y^T (I - H_X) Y + Y^T H_X Y$$

(b) $H_0: \beta_1 = \dots = \beta_p = 0$ vs. $H_1: \text{not } H_0$

$$F = \frac{SSR/p}{SSE/(n-p)} = \frac{Y^T H_X Y / p}{Y^T (I - H_X) Y / (n-p)}$$

$$\textcircled{1} \frac{Y^T H_X Y}{\sigma^2} \sim \chi^2(p, \frac{\beta^T X^T X \beta}{2\sigma^2})$$

$$\text{rank}(H_X) = \text{tr}(H_X) = p, \quad \frac{1}{2} \frac{\mu^T H_X \mu}{\sigma^2} = \frac{\beta^T X^T X \beta}{2\sigma^2}$$

idemp

$$\textcircled{2} \frac{Y^T(I-H)Y}{\sigma^2} \sim \chi^2(n-p)$$

$$\text{rank}(I-H) = \text{tr}(I-H) = n-p,$$

idemp

$$\frac{1}{2} \frac{\mu}{\sigma} (I-H) \frac{\mu}{\sigma} = \frac{1}{2} \frac{\mu}{\sigma} (X-X) \frac{\beta}{\sigma} = 0$$

$$\textcircled{3} \text{Independence} \quad HX(I-H) = HX - HX = 0$$

$$F = \frac{\frac{SSR}{\sigma^2} / p}{\frac{SSE}{\sigma^2} / (n-1)} \begin{matrix} \stackrel{H_1}{\sim} F(p, n-p, \frac{\beta^T X^T X \beta}{\sigma^2}) \\ \stackrel{H_0}{\sim} F(p, n-p) \end{matrix}$$



5.

$$\hat{e}^* = (I - H_{x_1})Y$$

$$\hat{e}^* = x_2 \beta_2^* + \gamma$$

$$\hat{\eta} = (I - H_{x_2})\hat{e}^*$$

$$= (I - H_{x_2})(I - H_{x_1})Y$$

$$\frac{\hat{\eta}^T \hat{\eta}}{n^2} = \frac{Y^T (I - H_{x_1})(I - H_{x_2})(I - H_{x_2})(I - H_{x_1})Y}{n^2}$$

$$= \frac{Y^T (I - H_{x_1})(I - H_{x_2})(I - H_{x_1})Y}{n^2}$$

$(I - H_{x_1})(I - H_{x_2})(I - H_{x_1})$ should be idempotent.

$$(I - H_{x_1})(I - H_{x_2})(I - H_{x_1})(I - H_{x_1})(I - H_{x_2})(I - H_{x_1})$$

$$= (I - H_{x_1})(I - H_{x_2})(I - H_{x_1})(I - H_{x_2})(I - H_{x_1})$$

should be $(I - H_{x_1})(I - H_{x_2})(I - H_{x_1})$

$$\text{If } X_1^T X_2 = 0,$$

$$(I - H_{x_1})(I - H_{x_2})(I - H_{x_1})(I - H_{x_2})(I - H_{x_1})$$

$$= (I - H_{x_1} - H_{x_2})(I - H_{x_1})(I - H_{x_1} - H_{x_2})$$

$$= (I - H_{x_1} - H_{x_2})(I - H_{x_1} - H_{x_2})$$

$$= I - H_{x_1} - H_{x_2}$$

$$\& (I - H_{x_1})(I - H_{x_2})(I - H_{x_1}) = I - H_{x_1} - H_{x_2}$$

$$\text{So, } (I - H_{X_1})(I - H_{X_2})(I - H_{X_1}) = I - H_{X_1} - H_{X_2}$$

I is idempotent under $X_1^T X_2 = 0$

$$\text{tr}(I - H_{X_1} - H_{X_2}) = n - p - q$$

$$(\beta_1^T X_1^T + \beta_2^T X_2^T)(I - H_{X_1} - H_{X_2})(X_1 \beta_1 + X_2 \beta_2) = 0$$

$$\text{So, } \frac{\hat{\eta}^T \hat{\eta}}{\sigma^2} \sim \chi^2(n - p - q)$$

□

6.

$$(a) \quad U = (y - X\beta)^T (y - X\beta) + \lambda^T (C\beta - 0)$$

$$\frac{\partial U}{\partial \lambda} = C\beta = 0$$

$$\frac{\partial U}{\partial \beta} = -2X^T y - 2X^T X \beta + C^T \lambda = 0$$

$$\Rightarrow \hat{\beta}_C = (X^T X)^{-1} X^T y - \frac{1}{2} (X^T X)^{-1} C^T \lambda$$

$$\text{Since } C\hat{\beta}_C = 0,$$

$$C(X^T X)^{-1} X^T y - \frac{1}{2} C(X^T X)^{-1} C^T \lambda = 0$$

$$\lambda = 2 [C(X^T X)^{-1} C^T]^{-1} C(X^T X)^{-1} X^T y$$

$$\begin{aligned} \hat{\beta}_C &= \hat{\beta} - \frac{1}{2} 2 \cdot (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} C(X^T X)^{-1} X^T y \\ &= \hat{\beta} - (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} C \hat{\beta} \end{aligned}$$

□

$$(b) \quad B := (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} C$$

$$Q\lambda = 0 \Rightarrow \lambda = 0$$

$$\text{Let } Q\lambda = 0 \Rightarrow C\lambda = 0 \text{ \& } P_1^T \lambda = 0$$

$$\text{If } P_1^T \lambda = 0 \Leftrightarrow \lambda \in N(P_1^T) \Leftrightarrow \lambda \in \mathcal{C}(P_2)$$

$$\Rightarrow \lambda = P_2 v \text{ for some } v$$

$$(I - B)P_2 = 0 \Rightarrow (I - B)P_2 v = (I - B)\lambda = 0$$

$$\lambda - B\lambda = 0 \Rightarrow \lambda = B\lambda$$

$$\Rightarrow \lambda = (X^T X)^{-1} C^T (C (X^T X)^{-1} C^T)^{-1} C \lambda$$

$$= 0$$

